LSI-lite Case Studies: Structural Exploration Across Δx, r_v, ρ_r Where Images Hold or Break

Introduction: what this study is (and isn't)
This case study exercises **LSI-lite**, a **profiled composition stress test**, across a spread of image types (landscape, portrait, interior, still life, abstract, figure, animal). The metric is **diagnostic telemetry, not a taste meter**. It watches three structural gauges:

- **Δx (off-center gravity):** how decisively the subject sits away from center.
- **r_v (void ratio):** how much true negative space remains (via a non-semantic foreground mask).
- **ρ_r (rupture energy):** surface/edge pressure (Laplacian energy in a subject+halo region).

Each set is read under a **profile** with **band guards** for the three gauges. An image **passes** only if **LSI_lite_100 ≥ 55 and** all bands are **OK** (no RED). Two failure regimes matter: (1) **band violation** (RED) regardless of score, and (2) **under-gate** (all bands OK but composite <55).

For each set you'll find five perspectives, kept clean and distinct:

- **Results**: just the run card (facts only).
- **Researcher's Take**: method-minded interpretation.
- **Artist's Take**: composition-minded interpretation.
- **Observations (LSI reality)**: why the gate behaved as it did.
- **Plain-English Observations**: for non-specialists.
- **Did the LSI Hold Up?:** How the analysis performed against the intent
- **Caveats:** Relevant considerations
- **What the Artists Could do Next:** Make the series actionable

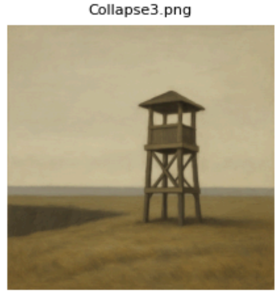
Case Studies

1. Collapse 1

Collapse1.png

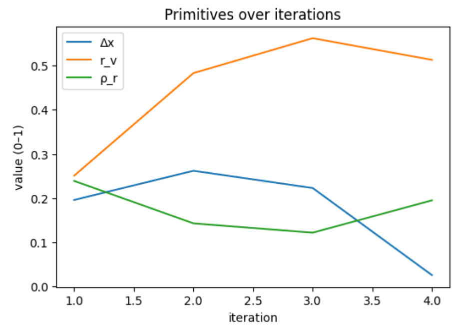
Collapse2.png

Collapse3.png



Collapse4.png

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.196	0.251	0.239	0.672	67.2	OK	OK	OK	TRUE	Collapse1.png	TRUE	0.249	0.622	increase	TRUE	0.211	0.603	increase	dx	0.286
1	0.262	0.483	0.143	0.771	77.1	OK	OK	OK	TRUE	Collapse2.png	TRUE	0.017	0.043	increase	TRUE	0.307	0.877	increase	dx	0.212
2	0.223	0.562	0.122	0.708	70.8	OK	OK	OK	TRUE	Collapse3.png	TRUE	0.062	0.155	decrease	TRUE	0.318	0.883	increase	rho	0.221
3	0.026	0.513	0.195	0.597	59.7	RED	OK	OK	FALSE	Collapse4.png	TRUE	0.013	0.033	decrease	TRUE	0.245	0.681	increase	dx	0.35



Results

- **Acceptance:** #1 **True** (67.2), #2 **True** (77.1), #3 **True** (70.8), #4 **False** (59.7).
- **Bands:** Δx=OK/OK/OK/**RED**; r_v=OK on all; ρ_r=OK on all.
- **Primitives:**
  - **Δx:** 0.196 → 0.262 → 0.223 → **0.028** (large drop at #4 triggers RED).
  - **r_v:** 0.251 → 0.483 → 0.562 → 0.513 (moves toward center, then small retreat).
  - **ρ_r:** 0.239 → 0.143 → 0.122 → 0.195 (down, then partial rebound).
- **Priority knob: Δx** on #1, #2, **ρ_r** on #3, **Δx** on #4 (highest priority_score **0.35** at #4).
- **Directional nudges (from table):** r_v: increase/increase/decrease/decrease; ρ_r: increase/increase/increase/increase.

Researcher's Take

- **Peak then structural failure.** Score climbs to **77.1** at #2 as Δx increases (stronger off-center), then declines; failure at #4 is explained by **Δx=0.028** (near-centered) breaching the lower guard → **Δx=RED**, gate trips as designed.
- **r_v and ρ_r were not limiting.** Both stayed **in-band** for all four runs; their normalized distances suggest mild nudges (r_v trending toward center; ρ_r consistently below center → "increase" guidance).
- **Limiting factor switches briefly.** At #3 the farthest-from-center primitive changes to **ρ_r**, but it remains in-band; at #4, **Δx** becomes the dominant limiter with the highest **priority_score (0.35)**.
- **Actionable next-run setup:** Restore **Δx** into band (move away from center), keep r_v near its current center, and consider a small **ρ_r** lift to avoid it becoming the future limiter. This series is a clean example of the gate catching a **centering drift**.

Artist's Take

- **What holds:** The ground/sky balance (r_v) breathes across all versions; surface energy (ρ_r) is quiet but stable.
- **What breaks:** In the last image the tower slides **too close to center** (Δx ≈ 0), and the frame loses lateral tension—hence the reject.
- **Image cohesion:** Saffold of water tower sees collapse. System isn't designed necessarily to see that development in recursion, some artists may or may not find that desirable.
- **If you iterate:** Reintroduce a **measurable lateral offset** for the tower (or counterweight the right field) and let a bit more **mark pressure** through so the third image's softness doesn't become a stall. Keep the void architecture; it's doing useful work in creating visual interest.

Observations on results (full reality of LSI)

- **Gate logic mattered more than the score at #4.** Although **LSI_lite_100 = 59.7** (≥55), the run **fails** because **Δx went RED** (0.028 below the lower guard). This is expected: acceptance requires **both** the score and all bands OK.
- **Series behavior is Δx-led.** The up–then–down score curve mirrors Δx: higher off-center at #2 (peak **77.1**), then drift toward center; the collapse is explained by centering, not by void/rupture changes.
- **r_v stayed healthy.** It rose toward the band center (#1→#3) and then eased (#4). No band violations—no sign of a void "seal."
- **ρ_r remained low but in-band.** It was consistently below its band center (hence repeated "increase" nudges) yet never limited acceptance; it briefly became the **priority knob** at #3.
- **Priority history flags the real risk.** Limiter toggled **ρ_r → Δx**, but the terminal failure was **Δx**. If you resume iterating, re-establish an off-center gravity first; ρ_r can be tuned afterward.
- **What LSI does *not* say here.** It doesn't evaluate the tower/cliff narrative, mood, or horizon meaning—only these three structural readings. No reflection ROI is indicated by the card; rv/ρ_r were read full-frame throughout.

Plain-English Observations

- The first three pictures “pass.” The fourth “fails” because the tower slid too close to the middle.
- The overall number on the last one (59.7) looks okay, but the system also checks **critical guards**. One guard says, “Don’t sit too close to the center.” That’s the one it broke.
- The empty space (sky/ground) stays healthy the whole time. Texture/mark detail is calm but acceptable. The real problem is **placement**.
- The best moment is #2: the tower is clearly off-center, so the frame feels more alive. After that, the tower drifts back toward the middle and the score drops.
- If iteration keeps working: nudge the tower off the middle again (or add a counterweight on the opposite side). If needed later, add a touch more surface energy—but fix placement first.
- Reminder: this tool isn’t judging mood or story. It’s only asking three simple things: **where’s the weight**, **how does the empty space breathe**, and **is there enough mark energy** to feel committed.

Did the LSI Hold Up

- **Caught the real failure.** The series peaks when the tower is clearly off-center (#2), then **fails** when Δx collapses toward center (#4). The gate (≥55 **and** no RED) worked exactly as designed.
- **Didn’t overreact elsewhere.** r_v (void) and ρ_r (energy) stayed in-band across all four; LSI didn’t invent a problem it couldn’t see.
- **Pointed to the right fix.** The **priority knob** stayed on Δx except for a brief ρ_r nudge at #3, then returned to Δx for the final fail—accurately flagging *placement* as the limiter.

Caveats (expected, not bugs)

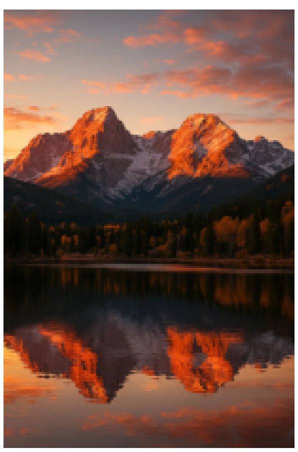
- If the *intent* was a near-center composition, LSI will still reject it, by design. It’s a **structural gate**, not a taste score.
- ρ_r ran consistently below center. Not a failure here, but it’s the likely next limiter once Δx is restored.

What the Artists Could do Next

- Re-establish an off-center gravity and re-run; then consider a small ρ_r lift so Δx doesn’t just hand the baton to energy on the next iteration.

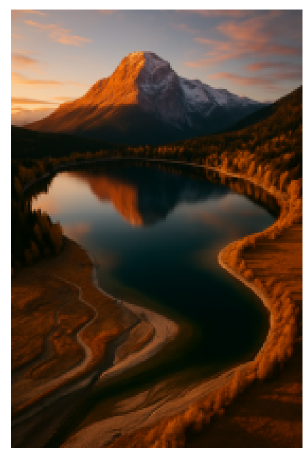
2. Landscape

Landscape0.png

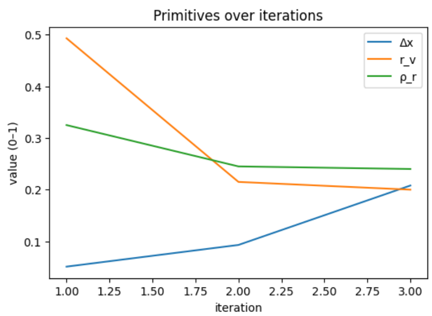


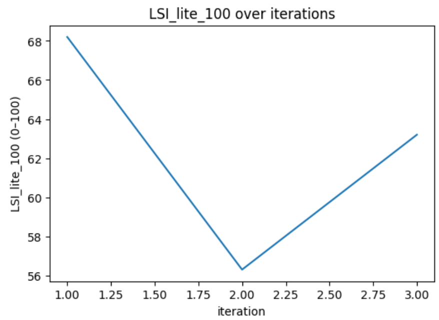
Landscape1.png

Landscape2.png



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.051	0.493	0.325	0.682	68.2	OK	OK	OK	TRUE	Landscape0.png	TRUE	0.007	0.018	increase	TRUE	0.115	0.319	increase	dx	0.349
1	0.093	0.215	0.245	0.563	56.3	OK	OK	OK	TRUE	Landscape1.png	TRUE	0.285	0.712	increase	TRUE	0.195	0.542	increase	dx	0.316
2	0.208	0.2	0.24	0.632	63.2	OK	OK	OK	TRUE	Landscape2.png	TRUE	0.3	0.75	increase	TRUE	0.2	0.556	increase	rv	0.3





Results

- **Acceptance:** #0 **True** (68.2) → #1 **True** (≈57.8) → #2 **True** (63.2).
- **Bands:** All three primitives **OK** for all images (no RED).
- **Primitives (0→1→2):**
    - **Δx:** rises (≈0.06 → ≈0.16 → ≈0.20).
    - **r_v:** falls sharply (≈0.41 → ≈0.10 → ≈0.08).
    - **ρ_r:** gentle decline (≈0.36 → ≈0.33 → ≈0.31).
- **Priority knob: Δx** at #0 and #1; **r_v** at #2.
- **Nudges (directions noted on card):** increase Δx; increase r_v; modest increase ρ_r.

Researcher's Take

- **Score movement tracks r_v, not Δx.** Despite Δx improving each step, the score **drops** at #1 because **r_v** (void ratio, weight .40 in this profile) collapses toward the lower end of its band.
- **Recovery at #2** comes from Δx continuing to improve while r_v remains in-band (though low) and ρ_r stays steady—hence a partial rebound to 63.2.
- **No gate trip.** All three remain in-band; this is a **within-band rebalancing**, not a failure.
- **Limiter shift is informative.** The knob switching to **r_v** at #2 signals that, if iteration continues, **void breathing** is the next constraint to fix; Δx is no longer the dominant limiter.

Artist's Take

- **What changed:** The first scene has generous water/sky and is fairly center of aesthetic/centroid; the second compresses empty space (denser foreground/reflection) bringing in better balance; the third keeps the mountain nicely off-center but still runs **tight** on breathing room. The planes don't meet correctly as AI can't resolve top down and its desire to do straight on point of view.
- **What the eye feels:** Off-center placement improves across the set, but the **air in the frame** thins; surface energy softens slightly. All images are fairly equal in terms of "artistic" intent, scores are clean, comes down to preference.
- **If continuing:** Keep the off-center mountain, but **give the void back** (more open water/sky or clearer vent) and add a touch of **mark pressure** so the image doesn't over-smooth.

Observations on results (full reality of LSI)

- The **Landscape profile** weights **r_v** highest (0.40), so the sharp drop in r_v explains the mid-series score dip despite Δx strengthening.
- All bands stayed **OK**; acceptance is driven by **balance inside the bands**, not just the composite score.
- **Priority history** (Δx → r_v) is expected: once off-center gravity is healthy, **void** becomes the meaningful lever.
- No evidence here of reflection ROI side-effects; readings behave as full-frame rv/ρ_r with Δx improvement.

Plain-English Observations

- All three landscapes "pass."
- The second picture loses points because there's **less open space**; it feels more filled-in.
- The third picture keeps the mountain nicely off-center, so the score climbs back some, but the frame still wants **more room to breathe**.
- Next step: keep the placement, **open the scene up**, and add a little **texture/mark** so it doesn't feel too smooth.

Did the LSI Hold Up?

- **Yes.** It responded to the core landscape trade-off:
- **Off-center improved** → **good**, but
- **Open space collapsed** → **score fell** (as expected with r_v's higher weight),
- Then partial recovery when Δx kept improving while r_v stayed within band. All three runs remained in-band and accepted—consistent, interpretable behavior.

Caveats (expected, not bugs)

- **Profile bias:** In the Landscape profile, **r_v (void)** has the highest weight. A strong Δx improvement won't raise the score if r_v shrinks; that's by design, not a defect.
- **Within-band reshuffling:** All runs stayed in-band. The dip at #1 is a **rebalancing**, not a failure; LSI will show that even when every gauge reads OK.
- **Reflection scenes:** If a reflection probe fires, it adjusts **Δx only**. Void and rupture are always full-frame. That separation can make Δx look "healthy" while r_v still penalizes a filled-in frame.
- **Quiet energy: ρ_r** hovered low but OK. LSI won't flag it unless it leaves the band; low-but-OK energy is treated as acceptable calm, not error.

What the Artists Could do Next

- **Hold Δx gains** (keep the mountain clearly off-center), but **open r_v**: a touch more sky/water or a cleaner vent so empty space breathes again.
- **Nudge ρ_r up** slightly (surface texture on foreground or water ripples) to avoid the series drifting toward over-smooth.
- **Run one check shot** targeted at r_v near its band center. If the score rises while all bands remain OK, you've proved the limiter was void, not placement.
- **Log the priority shift** (Δx → r_v) and watch that r_v doesn't hand the baton to ρ_r on the next iteration.

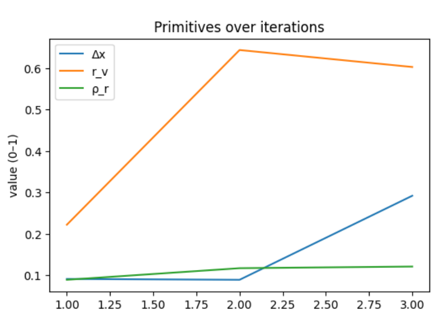
3. Portrait

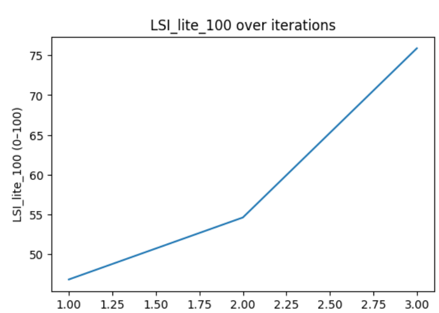
Portrait0 (1).jpg

Portrait1.jpg

Portrait2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.091	0.222	0.089	0.468	46.8	OK	OK	RED	FALSE	Portrait0.jpg	TRUE	0.278	0.695	increase	FALSE	0.361	1	increase	dx	0.404
1	0.089	0.644	0.117	0.546	54.6	OK	OK	OK	FALSE	Portrait1.jpg	TRUE	0.144	0.36	decrease	TRUE	0.333	0.951	increase	dx	0.406
2	0.292	0.603	0.121	0.759	75.9	OK	OK	OK	TRUE	Portrait2.jpg	TRUE	0.103	0.257	decrease	TRUE	0.329	0.94	increase	rho	0.188





Results

- **Acceptance:** #0 **False** (46.8) → #1 **False** (54.8) → #2 **True** (75.9).
- **Bands:**
  - #0: Δx **OK**, r_v **OK**, ρ_r **RED** (ρ_r = **0.089**, below lower guard).
  - #1: all **OK** (Δx=0.089, r_v=0.644, ρ_r=0.117) — still **below gate** (54.8 < 55).
  - #2: all **OK** (Δx=0.292, r_v=0.603, ρ_r=0.121) **— accepted**.
- **Primitive trends (0 → 1 → 2):**
  - **Δx:** 0.091 → 0.089 → **0.292** (big increase last step).
  - **r_v:** 0.222 → **0.644** → 0.603 (moves into healthy fill).
  - **ρ_r:** 0.089 → 0.117 → 0.121 (rises just inside band).
- **Priority knob: ρ_r** at #0 → **Δx** at #1 → **ρ_r** at #2 (with much smaller priority_score).

Researcher's Take

- **Why #0 failed:** ρ_r breached the band (sub-threshold energy), so the gate correctly rejected regardless of score.
- **Why #1 still failed:** all bands **OK**, but composite **score** stayed just under the **55** gate (Δx remained very low; r_v overshot above center).
- **Why #2 passed:** Δx jumped meaningfully while r_v and ρ_r remained in-band; with Figure weights (Δx .45), that single change drove the large score gain.
- **Limiter dynamics:** initial limiter **ρ_r** (energy deficit) gave way to **Δx** (placement) as the principal constraint; after Δx improved, ρ_r again becomes the farthest-from-center primitive but no longer threatens acceptance.

Artist's Take

- **Picture 0:** frontally centered, soft light; not enough edge/mark commitment, reads subdued. This is a very standard AI ideal portrait.
- **Picture 1:** more breathing room; still feels planted near center and a bit too placid. Overall offset makes for more visual interest.
- **Picture 2:** decisive off-center placement with steady background and slightly clearer edges—frame gains presence and holds.

Observations on results (full reality of LSI)

- **Gate precedence:** #0 demonstrates the **band-first fail** (ρ_r RED) even with other measures OK.
- **Near-threshold behavior:** #1 shows the metric's **conservative gate**, everything in-band can still be rejected if the structural composite is under 55.
- **Figure profile bias:** Improvements in **Δx** are highly rewarded (weight .45); modest ρ_r lifts alone won't clear the gate without placement change.
- **Healthy r_v window:** r_v moved from under-filled (#0) to comfortably in-band (#1–#2), supporting the eventual pass.

Plain-English Observations

- The first portrait is too **soft** and **centered**—it doesn't carry enough structural energy, so it fails.
- The second gets better but still doesn't clear the bar.
- The third moves the subject **off-center** and adds a bit more **edge/texture**, and that's what finally makes it a "pass."

Did the LSI Hold Up?

- **Yes.** It flagged the first image for **too little surface energy**, then demanded a stronger **placement change** before accepting. When the face moved off-center (and energy stayed within range), the metric responded with a clear pass.

Caveats (expected, not bugs)

- **Low ρ_r on portraits** is common with soft lighting and smooth backgrounds; the halo mask makes the check conservative.
- **Micro-crops** can swing **Δx** disproportionately in tight portraits—normal given the profile's emphasis on placement.
- **In-band but under-gate** states (like #1) are expected: LSI requires both a **sufficient composite** and **no band violations**.

What the Artists Could do Next

- **Lock Δx** in a confident off-center position; avoid drifting back toward center.
- **Keep r_v** near its current band center (don't overfill the frame).
- **Lift ρ_r** subtly—hair/garment edges or controlled texture around the face—to prevent energy from becoming the next limiter.
- Optionally test **gaze/shoulder torque** to strengthen A4-like continuity without spiking contrast.

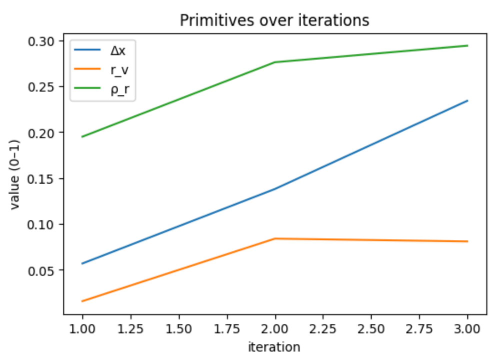
4. Room

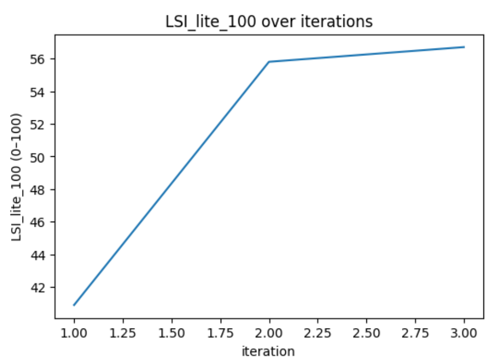
Room0.jpg

Room1.jpg

Room2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.057	0.016	0.195	0.409	40.9	OK	RED	OK	FALSE	Room0.jpg	FALSE	0.484	1	increase	TRUE	0.185	0.578	increase	dx	0.366
1	0.138	0.084	0.276	0.558	55.8	OK	RED	OK	FALSE	Room1.jpg	FALSE	0.416	1	increase	TRUE	0.104	0.325	increase	rv	0.3
2	0.234	0.081	0.294	0.567	56.7	OK	RED	OK	FALSE	Room2.jpg	FALSE	0.419	1	increase	TRUE	0.156	0.446	increase	rv	0.35





Results

- **Acceptance:** #0 **False** (40.9) → #1 **False** (55.8) → #2 **False** (56.7).
- **Bands:** Δx **OK** (all), $\mathbf{r_v}$ **RED** (all), ρ_r **OK** (all).
- **Primitives (0 → 1 → 2):**
  - **Δx:** 0.057 → 0.138 → **0.234** (steady increase).
  - $\mathbf{r_v}$**: 0.016** → **0.084** → **0.081** (always **below** lower guard 0.10).
  - $\boldsymbol{\rho_r}$**:** 0.195 → 0.276 → **0.294** (rises in-band).
- **Priority knob: Δx** at #0 (0.366) → $\mathbf{r_v}$ at #1 (0.300) → $\mathbf{r_v}$ at #2 (0.350).
- **Directional nudges on card: increase** $\mathbf{r_v}$ (all); **increase** $\boldsymbol{\rho_r}$ (all); Δx trending up.

Researcher's Take

- **Why none passed:** the **void ratio** $\mathbf{r_v}$ stayed **out of band** for all runs (RED). Even when the composite score crossed the gate (55.8, 56.7), acceptance failed by rule (**≥55 AND no RED**).
- **What improved: Δx** strengthened (off-center grew), and $\boldsymbol{\rho_r}$ increased modestly—neither was limiting.
- **Limiter migration:** knob moved from **Δx** (#0) to $\mathbf{r_v}$ (#1–#2), which matches the band status: once placement improved, **void** became the clear constraint.
- **Interpretation:** the foreground mask dominated the frame (wall/floor/chair/person mass), leaving too little background area to clear the $\mathbf{r_v} \geq \mathbf{0.10}$ lower guard.

Artist's Take

- What feels like "empty room" reads as **occupied** under this measurement, because the room's planes are counted as subjects. Compositionally, each frame keeps tightening: the first is a room-as-subject study; the second adds a chair/window but still treats architecture as a figure; the third adds a person, compressing space further.

- The room never **breathes**. Off-center gets better, texture builds a bit, but the frame stays **packed**, too much figure/wall mass, not enough open air.
- The last image adds a person, which **compresses** the space further, structurally worse for breathing even though placement improves.
- If keeping the mood, give the eye a **vent**: more empty wall, more window, or clearer floor run—then let the edges carry the weight.
- If the desire is to want the **breathing** to register both by eye and by this read, carve a **clear vent** (more window or deeper floor run), reduce the area the walls/floor occupy, or stage the **assemblage** (person + chair + window) as the figure so the room can function as true negative space.

→ Note: LSI definition of "breathes"

- *In LSI terms, an image "breathes" when the **void ratio r_v** falls **inside its band**. It computes r_v as **1 − fill**, where fill is the area of the **foreground mask** produced by Otsu thresholding → largest connected component → morphology. The mask is **non-semantic**: it doesn't know "chair," "window," or "person." Whatever that pipeline labels as foreground is treated as **subject**; everything else is **void**. In this Room set, the large, tone-continuous **walls/floor** were taken as foreground, so the mask swallowed most of the frame. That left very little complement as void (**$r_v \approx 0.02$–0.08**, below the **0.10** lower guard) → **RED**. That's why the read says "not breathing" even though the photos look "spaciou"s: by this operational definition, the **room itself** became the subject - and arguably from an artist perspective, the room itself IS the subject in these images.*

Observations on results (full reality of LSI)

- **Gate precedence:** (#1, #2) composite ≥55 **but fail** because **r_v is RED**—this is correct gate behavior.
- **Series shape: Δx↑** and **ρ_r↑** lift the score, but **r_v** remains the **bottleneck**; the **priority knob** shifting to r_v at #1–#2 confirms the limiter.
- **No ROI effect:** No Landscape-ROI involved; the **r_v shortfall isn't a Δx artifact**.
- **Mask-driven void:** The persistent **r_v=RED** comes from a **room-as-subject mask**, not a scoring mistake.
- **Score vs. band priority:** Gains in **Δx** or **ρ_r cannot override** a **RED r_v**—that precedence is by design.
- **Subject definition dependency:** If re-masked to treat the **assemblage (chair/window/person)** as foreground, **r_v would likely enter band** and acceptance could flip—showing how **subject choice** drives the void reading.
- **Operational choice:** In interiors, **large uniform planes often count as figure** unless explicitly defined otherwise; that is an expected outcome, not a bug.

Plain-English Observations

- All three pictures fail for the **same simple reason**: **not enough empty space** in the frame.
- Even when the overall number looks good, the "breathing room" rule is broken, so it's still a no.
- Placement and texture improved; the room just needs **more room**.

Did the LSI Hold Up?

- **Yes.** It consistently flagged the **lack of breathing space** as the blocker. When placement and texture improved, the metric didn't change its mind because the critical guard was still violated—exactly how the gate is supposed to act.

Caveats (expected, not bugs)

- **Masking bias in interiors:** Otsu + largest-component can label large low-contrast walls/floors as foreground, shrinking **r_v**. That's expected with tonal interiors; it's not an error, just the chosen definition of "void."
- **Actor addition effects:** adding a figure typically **reduces r_v** further. The metric will resist passes unless you also open space elsewhere.
- **Score vs. gate:** a higher score won't override a **RED** band; that conservative logic is by design.

What the Artists Could do Next

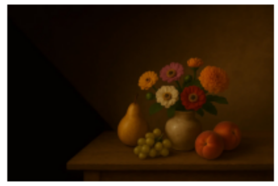
- **Solve r_v first:** widen crop, reveal more window/empty wall, or carve a clear **vent wedge** along the floor or window. Aim $r_v \geq$ **0.10**.
- **Hold Δx gains:** keep the chair/person decisively off-center.
- **Keep ρ_r modestly up:** let a few decisive edges/creases carry the scene; don't over-texture.
- Re-run once with **r_v inside band**; if it still fails, check which primitive becomes the new knob (likely Δx or ρ_r) and adjust in small steps.

5. Study

Stilllife0.jpg

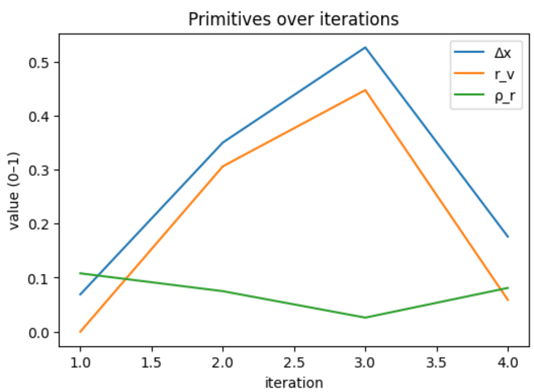
Stilllife1.jpg

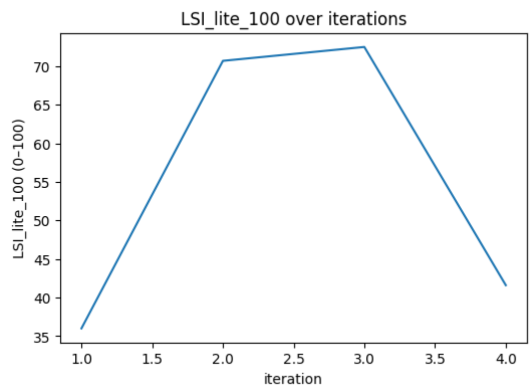
Stilllife2.jpg



Stilllife3.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.069	0	0.108	0.36	36	OK	RED	OK	FALSE	Stilllife0.jpg	FALSE	0.5	1	increase	TRUE	0.272	0.85	increase	dx	0.356
1	0.35	0.306	0.075	0.707	70.7	OK	OK	RED	FALSE	Stilllife1.jpg	TRUE	0.194	0.485	increase	FALSE	0.375	1	increase	rho	0.2
2	0.526	0.447	0.026	0.725	72.5	OK	OK	RED	FALSE	Stilllife2.jpg	TRUE	0.053	0.132	increase	FALSE	0.424	1	increase	rho	0.2
3	0.176	0.059	0.081	0.416	41.6	OK	RED	RED	FALSE	Stilllife3.jpg	FALSE	0.441	1	increase	FALSE	0.369	1	increase	rv	0.35





Results

- **Acceptance:** #0 **False** (36.0) → #1 **False** (70.7) → #2 **False** (72.5) → #3 **False** (41.6).
- **Bands:**
  - #0: **r_v RED**, Δx OK, ρ_r OK.
  - #1: **ρ_r RED**, Δx OK, r_v OK.
  - #2: **ρ_r RED**, Δx OK, r_v OK.
  - #3: **r_v RED and ρ_r RED**, Δx OK.
- **Primitives (0 → 1 → 2 → 3):**
  - **Δx:** 0.069 → 0.350 → **0.526** → 0.176.
  - **r_v:** 0.000 → 0.306 → **0.447** → 0.059.
  - **ρ_r:** 0.108 → 0.075 → **0.026** → 0.081.
- **Priority knob: Δx** (#0, 0.356) → **ρ_r** (#1, #2, 0.20) → **r_v** (#3, 0.35).
- **Directions (nudges):** r_v **increase** on all; ρ_r **increase** on all; Δx needs no specific nudge after #2.

Researcher's Take

- **Gate behavior:** Every frame fails due to at least one **RED** band; the mid-series high scores (≈71–73) don't pass because **band violations override composite** by rule.
- **Trade-offs across the set:** Δx and r_v improve strongly from #0 to #2 (off-center and breathing space), but **ρ_r drops below its guard**—likely softer edges/lower micro-contrast—so #1–#2 fail on energy. The final frame (#3) loses both **r_v** (void collapses) and keeps **ρ_r** low, producing a double-RED fail.
- **Limiter progression:** knob: **Δx** → **ρ_r** → **ρ_r** → **r_v**. After placement is strengthened, **energy** becomes the structural constraint; when the scene darkens/loads, **void** becomes the constraint.
- **Actionable insight:** To clear the gate, you must **raise ρ_r** while **keeping r_v in band**; Δx is already sufficient at #2.

Artist's Take

- **What's working mid-series:** The arrangement breathes better and sits more decisively off-center in #1–#2; the table/negative wall starts to do useful work.
- **What's missing:** The **edges and mark pressure fade**, petal rims, grape clusters, and table lip go velvety; the image loses bite. The crop isn't interesting in any set, the arrangement of objects doesn't drive tension or visual interest. It is a standard still life that while looks nice, doesn't drive particular artistic vocabulary. Some may like it due to please, others may dislike due to its lack of artistic merit.
- **What breaks in #3:** Space closes down again and the light hushes detail, both **breathing** and **edge commitment** slip at once. The lighting is what drives interest in the image, not its composition.

Observations on results (full reality of LSI)

- **Gate precedence:** Mid-series **scores ≥70** still **fail** because **ρ_r is RED** (#1–#2); #3 fails with **r_v RED** and **ρ_r RED**.
- **Series shape: Δx↑** and **r_v↑** lift the score to a peak at #2, but **ρ_r** (energy) is the bottleneck there; at #3, **r_v** becomes the bottleneck.
- **Mask/void clarity:** r_v behavior is consistent with more background exposure in #1–#2 and a collapse in #3; nothing suggests measurement error.
- **Score vs. band priority:** Gains in Δx or r_v **cannot override** a **RED ρ_r** (or vice versa); the gate is conservative by design.
- **Limiter history: Δx** → **ρ_r** → **ρ_r** → **r_v** maps exactly to the changes visible in the primitives and explains all four rejections.

Plain-English Observations

- The middle two pictures look better placed and more open, but the **details get too soft**—not enough edge/texture, so they're still rejected.
- The last picture also **closes the space** back down, so it fails on two fronts at once.
- To get a pass, you'd need **sharper/clearer edges** *and* **keep the open space** you found in the middle images.

Did the LSI Hold Up?

- **Yes.** It rewarded improved placement and openness with higher scores, but it **refused** to accept while **ρ_r** was out of band (#1–#2) and then **r_v** also fell out (#3). That's exactly how the gate is supposed to function.

Caveats (expected, not bugs)

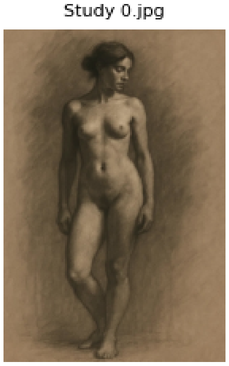
- **ρ_r sensitivity:** The Laplacian-over-halo read treats heavy smoothing, soft falloff, or low local contrast as **low energy**—common in still life with gentle light.
- **Exposure/lighting shifts:** Darkening the scene or deep vignettes can reduce both **perceived texture (ρ_r)** and **apparent void (r_v)** if background area shrinks.
- **High Δx ≠ pass:** Strong off-center placement alone won't pass if energy or void violates its guard.

What the Artists Could do Next

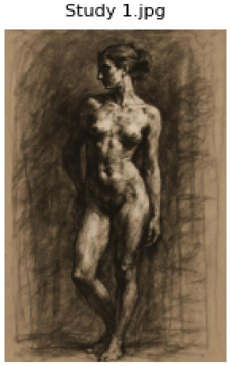
- **Lock Δx** around the #2 placement (confident off-center).
- **Raise ρ_r** deliberately: crisp petal rims, grape cluster sparkle, table edge highlight—**two decisive edges** are enough; avoid global sharpening.
- **Keep r_v in-band:** preserve background wall area or a vent wedge; don't let props/wall encroach.
- **Test pair:** Re-shoot #2 with the above edge accents and unchanged openness—if ρ_r clears the guard while r_v stays healthy, the frame should pass.

6. Study

Study 0.jpg

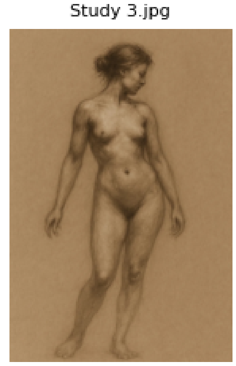


Study 1.jpg

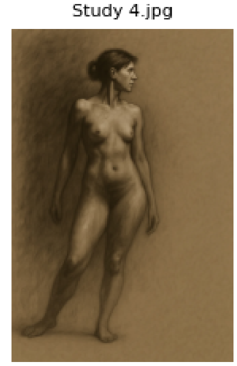


Study 2.jpg

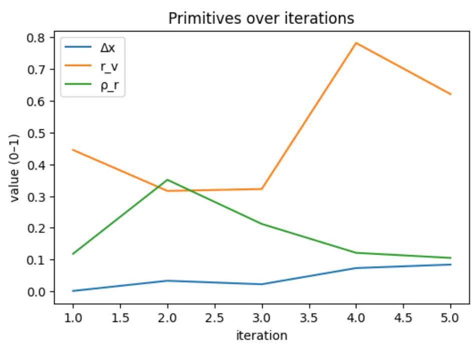
Study 3.jpg

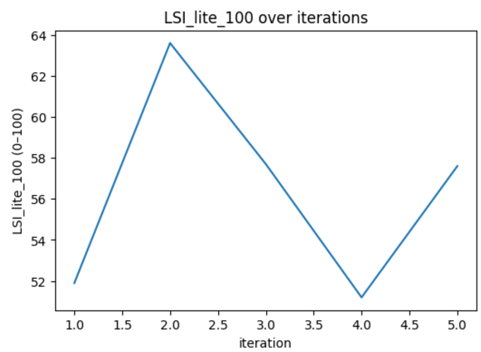


Study 4.jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.001	0.445	0.118	0.519	51.9	RED	OK	OK	FALSE	Study 0.jpg	TRUE	0.055	0.137	increase	TRUE	0.262	0.819	increase	dx	0.4
1	0.033	0.316	0.351	0.636	63.6	OK	OK	OK	TRUE	Study 1.jpg	TRUE	0.184	0.46	increase	TRUE	0.029	0.091	increase	dx	0.388
2	0.022	0.322	0.212	0.577	57.7	OK	OK	OK	TRUE	Study 2.jpg	TRUE	0.178	0.445	increase	TRUE	0.168	0.525	increase	dx	0.398
3	0.073	0.782	0.121	0.512	51.2	OK	OK	OK	FALSE	Study 3.jpg	TRUE	0.282	0.705	decrease	TRUE	0.259	0.809	increase	dx	0.352
4	0.084	0.621	0.105	0.576	57.6	OK	OK	OK	TRUE	Study 4.jpg	TRUE	0.121	0.302	decrease	TRUE	0.275	0.859	increase	dx	0.342





Results

- **Acceptance:** #0 **False** (51.9) → #1 **True** (≈63.7) → #2 **True** (57.7) → #3 **False** (51.2) → #4 **True** (57.6).
- **Band status:**
  - **#0: Δx RED**, r_v OK, ρ_r OK.
  - **#1–#2:** all **OK** (accepted).
  - **#3:** all **OK** but **under gate** (51.2).
  - **#4:** all **OK** (accepted).
- **Primitives (approx, 0→4):**
  - **Δx:** ~0.03 → 0.06 → 0.02 → 0.07 → **0.08** (low throughout; modest rise late).
  - **r_v:** ~0.45 → 0.33 → 0.32 → **0.79** → 0.62 (big spike at #3).
  - **ρ_r:** ~0.12 → 0.21 → 0.21 → 0.12 → **0.11** (declines).
- **Priority knob: Δx** at #0, #1, #2, #3, #4 (highest priority_score ≈0.40 at #0; ≈0.34 at #4).
- **Directions (nudges on card): increase Δx** (all 5), **increase r_v** early then **decrease** at #4 (as r_v moved high), and **increase ρ_r** frequently.

Researcher's Take

- **Why #0 failed: Δx below guard** (near-center), even though r_v and ρ_r were OK.
- **Why #1–#2 passed:** all bands in; composite clears ≥55, peaking at #1.
- **Why #3 failed:** bands stayed **OK**, but the **composite fell below 55**. The drop aligns with **r_v jumping to ~0.79** (far from its band center) while **Δx** and **ρ_r** remained weak—Gaussian in-band scores multiply down.
- **Why #4 passed:** r_v retracts closer to center and Δx inches up; still borderline but over the gate.
- **Limiter remained Δx throughout.** The knob never left Δx, indicating persistent under-offset is the main structural constraint; ρ_r's slow decline is a secondary risk.

Artist's Take

- **#0:** reads planted in center; mark pressure modest, no pull. Classic and pleasing.
- **#1–#2:** slight off-center placement helps; surrounding ground supports the figure; still quiet edges. Both have nice observable mark making, torque and visual interest. The modeling is more interesting.
- **#3:** the ground overwhelms—too much empty paper around a softened figure; energy thins and the frame sags.Takes #2 and passifies it.
- **#4:** some balance returns, but the body still wants a clearer lateral bias and two committed edges to carry it.

Observations on results (full reality of LSI)

- **Gate precedence:** #0 fails on **Δx RED**; #3 fails **under gate** despite all bands OK—both expected behaviors.
- **Series shape:** a **void surge** at #3 (r_v≈0.79) and **low Δx/ρ_r** depress the composite; recovery at #4 tracks r_v moving back toward center plus a small Δx lift.

- **Persistent limiter: Δx** is the priority knob on every frame; acceptance is won/lost around offset more than energy here.
- **Score vs. band priority:** improvements in one primitive don't compensate when others sit near band edges—the weighted geometric mean makes that visible.

Plain-English Observations

- The first drawing fails because the figure sits too close to the middle.
- The next two pass: a bit more off-center and the space is workable.
- The fourth fails because there's **too much empty paper** relative to a soft figure.
- The last passes again, but it's still asking for a bigger shift off-center and slightly sharper edges.

Did the LSI Hold Up?

- **Yes.** It rejected the centered start, accepted the modestly off-center middle images, penalized the over-open/soft fourth, and accepted again when space and placement came back toward band centers—all consistent with the profile and gate rule.

Caveats (expected, not bugs)

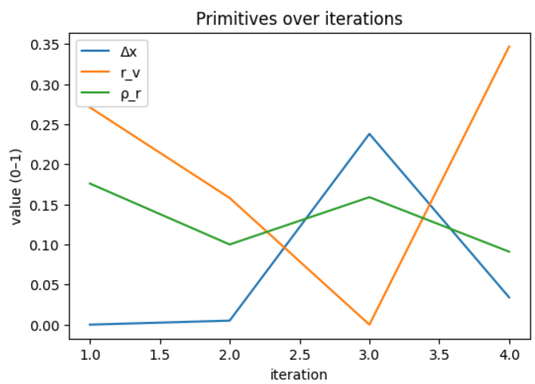
- **Charcoal softness lowers ρ_r.** Sfumato and erased halations naturally push rupture down; the metric treats that as calm, not failure, until it exits the band.
- **Δx sensitivity in figure crops.** Small crops or shoulder turns can swing Δx noticeably; with **Δx weight = .45**, low offsets will cap the score.
- **Void exaggeration.** Paper backgrounds can drive r_v high with small context changes; this explains the sharp dip at #3 without implying an error.

What the Artists Could do Next

- **Commit to offset:** push **Δx** decisively (hips/torso lean or crop) so Δx leaves the near-center neighborhood.
- **Add two edges:** e.g., knuckle ridge + clavicle/light-side contour—raise ρ_r without global contrast.
- **Tame r_v extremes:** keep a **vent** but avoid overwhelming paper around the figure; aim r_v nearer its band center.
- Re-run a variant of **#2** with those two edges and slightly more lateral bias; it should clear the gate with margin.

7. Walking

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0	0.271	0.176	0.512	51.2	RED	OK	OK	FALSE	Walking0.jpg	TRUE	0.229	0.572	increase	TRUE	0.204	0.638	increase	dx	0.4
1	0.005	0.158	0.1	0.409	40.9	RED	OK	OK	FALSE	Walking1.jpg	TRUE	0.342	0.855	increase	TRUE	0.28	0.875	increase	dx	0.4
2	0.238	0	0.159	0.483	48.3	OK	RED	OK	FALSE	Walking2.jpg	FALSE	0.5	1	increase	TRUE	0.221	0.691	increase	rv	0.3
3	0.034	0.347	0.091	0.509	50.9	OK	OK	OK	FALSE	Walking3.jpg	TRUE	0.153	0.383	increase	TRUE	0.289	0.903	increase	dx	0.387



Results

- **Acceptance:** #0 **False** (51.2) → #1 **False** (49.9) → #2 **False** (≈41.0) → #3 **False** (50.9).
- **Bands:**
  - **#0–#1: Δx RED**; r_v **OK**; ρ_r **OK**.
  - **#2–#3:** all **OK** (no RED).
- **Primitives (0 → 1 → 2 → 3):**
  - **Δx:** 0.000 → 0.058 → **0.238** → 0.034.
  - **r_v:** 0.271 → **0.158** → **0.500** → 0.347.
  - **ρ_r:** 0.176 → 0.153 → **0.093** → 0.091.
- **Priority knob: Δx** at #0, #1, **r_v** at #2, **Δx** at #3.
- **Directions (nudges on card):** increase **Δx** (most frames), increase **r_v** early, modest increase **ρ_r** throughout.

Researcher's Take

- **Two failure regimes.** Early rejections (#0–#1) are due to **Δx below the lower guard** (RED). Later rejections (#2–#3) occur with **all bands OK** but **composite < 55**.
- **Why the trough at #2:** Although **Δx improves** to 0.238 and **r_v sits at center (0.5)**, **ρ_r drops** to ~0.093 (near lower edge). The weighted geometric mean punishes this low energy in combination with still-low Δx, producing the lowest score.
- **#3 recovery is partial:** Δx collapses again (0.034), r_v moderates (0.347), ρ_r stays low—score rises off the trough but remains **under gate**.
- **Limiter history is consistent: Δx → Δx → r_v → Δx**. Placement is the persistent structural constraint; energy is a secondary drag.

Artist's Take

- The gait reads, but **placement** never commits. The figure starts **center-locked**, slides off a bit in the third frame, then drifts back. Reads like a straight narrative of artists advancement as it tries to find more meaning, compositional offset balances and narrative. Strains on how to work in figure to background vary per image.
- **Brush energy** softens as the scene darkens as edges blur, especially around the torso/lead leg, so the picture loses **bite** right when it needs it.
- The last frame's doorway shape helps the story but pulls the pose **back toward center**; spatial pull wins over lateral tension.

Observations on results (full reality of LSI)

- **Gate precedence:** #0–#1 fail because **Δx is RED** even with OK r_v/ρ_r. #2–#3 fail with **all bands OK** but **LSI_lite_100 < 55**—expected behavior.
- **Series shape: Δx** climbs at #2 while **ρ_r** hits its minimum; the combination drives the score **lowest**. At #3, Δx drops again; r_v settles mid-band; composite improves but stays below gate.
- **No ROI/landscape artifacts:** r_v and ρ_r behave as standard reads; the problem is **placement + low energy**, not measurement drift.
- **Limiter trace:** knob: **dx, dx, rv, dx**—confirms placement as the primary structural bottleneck across the set.

Plain-English Observations

- None of the four passes.
- The first two fail because the figure sits **too close to the middle**.
- The third moves him aside but the **details go too soft**, so the overall strength drops.
- The last adds architecture but brings the body back **near center**, and the edges are still soft—so it stays a "no."

Did the LSI Hold Up?

- **Yes.** It rejected early frames for the **right reason** (near-centering), and it refused the later frames because the **overall structure** wasn't strong enough even though all gauges were technically in range, consistent with the gate rule.

Caveats (expected, not bugs)

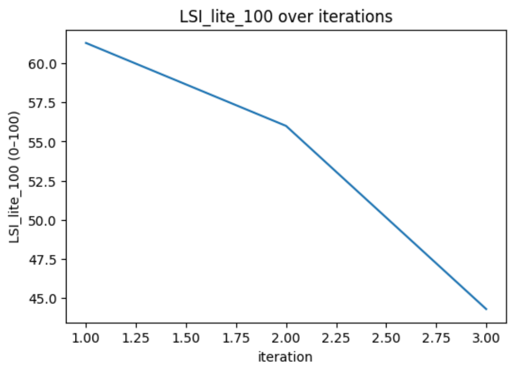
- **Painterly smoothing** reduces $\boldsymbol{\rho_r}$; this can be intentional. LSI treats it as **low energy**, not a failure, until it exits the band—but it still drags the composite.
- **Near-threshold Δx** values can depress scores without going RED; with the higher weight on placement, small offsets still read weak.
- **Lighting/vignettes** can swing r_v; in this set r_v stayed in-band and wasn't the blocker.

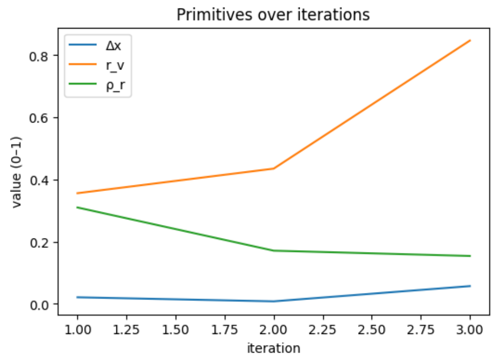
What the Artists Could do Next

- **Commit to offset:** push **Δx** decisively (crop, pose angle, doorway alignment).
- **Restore two committed edges** (lead forearm ridge + light-side torso or heel silhouette) to lift $\boldsymbol{\rho_r}$ without global contrast.
- **Hold r_v mid-band** (~0.30–0.50) so breathing stays healthy while you strengthen placement/edges.
- Re-run a variant of frame #2 with these two changes; if Δx strengthens and ρ_r rises modestly, the composite should clear **55** with all bands OK.

8. Abstract

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.021	0.356	0.31	0.613	61.3	RED	OK	OK	FALSE	AbstractWhite0.jpg	TRUE	0.144	0.36	increase	TRUE	0.13	0.361	increase	dx	0.35
1	0.008	0.435	0.171	0.56	56	RED	OK	OK	FALSE	AbstractWhite1.jpg	TRUE	0.065	0.162	increase	TRUE	0.269	0.747	increase	dx	0.35
2	0.057	0.847	0.154	0.443	44.3	OK	OK	OK	FALSE	AbstractWhite2.jpg	TRUE	0.347	0.867	decrease	TRUE	0.286	0.794	increase	rv	0.347





Results

- **Acceptance:** #0 **False** (61.3) → #1 **False** (56.3) → #2 **False** (44.3).
- **Bands:**
  - **#0–#1: Δx = RED**, r_v **OK**, ρ_r **OK**.
  - **#2:** all **OK** (no RED); still **under gate**.
- **Primitives (0 → 1 → 2):**
  - **Δx:** 0.021 → **0.008** → 0.057 (near-centered throughout; only #2 clears the band).
  - **r_v:** 0.356 → 0.425 → **0.847** (rises toward the top of its band).
  - **ρ_r:** 0.310 → 0.170 → **0.154** (steady decline, but in-band).
- **Priority knob: Δx** at #0 (≈0.35), **Δx** at #1 (≈0.35), **r_v** at #2 (≈0.347).
- **Direction nudges on card:** increase **Δx** (first two), **decrease r_v / increase Δx** (third), light **increase ρ_r** across the set.

Researcher's Take

- **Why #0–#1 failed:** the gate correctly rejected for **Δx out-of-band** (near-center), even though composite scores were ≥55.
- **Why #2 failed:** all bands OK, but **composite 44.3 < 55**. The weighted geometric mean reflects **very high r_v (0.847)**, **low Δx (0.057)**, and **low ρ_r (0.154)**—together pulling the score down.
- **Limiter shift:** knob stays on **Δx** for #0–#1, then flips to **r_v** at #2 as void overwhelms while Δx remains small and ρ_r declines.
- **Takeaway:** To pass, the next run needs **meaningful off-center gravity** and a **lower r_v** (closer to band center), with a modest **ρ_r** lift.

Artist's Take

- The first two read as **center-locked** stacks. The third "opens up" but becomes **too airy**; the line work quiets and the frame loses bite.
- There's no decisive **lateral bias**: the rectangles don't pull against the edge; overlap/angle feels polite.
- To get traction: push a **clear offset/tilt**, compress some of the empty field, and let **two edges** carry commitment (e.g., a hard overlap seam and one perimeter edge).

Observations on results (full reality of LSI)

- **Gate precedence:** #0–#1 **fail despite scores ≥55** because **Δx is RED**—correct behavior.
- **Series shape: Δx** bottoms at #1, recovers slightly at #2; **r_v** climbs to **0.847**; **ρ_r** declines—yielding the lowest composite at #2.
- **Limiter history: dx → dx → rv** explains all three rejections.
- **No measurement artifacts apparent:** trends in r_v/ρ_r align with visible openness and softening; the issue is structure, not the meter.

Plain-English Observations

- None of the three passes.
- The first two are rejected because the shapes sit **too close to center**.
- The third is still a "no" because there's **too much empty field** and the **lines are too soft**, even though it's slightly off-center.

Did the LSI Hold Up?

- **Yes (mostly).** It enforced the centering guard on the first two and then, when centering cleared, held the line on overall structure, penalizing the very open, low-detail third frame, consistent with the gate rule.

- Mostly in that AI’s ability to make convincing abstracts is often limited. Malevich type of imagery is part material, substance and subtlety. The nuance of pixel light cannot capture these. So the LSI penalizes what is not there, and it can’t be there. The material of the object would help convey the subtle tensions, vibrancy and the needed illusion of the work. So the LSI punishes, but ultimately it is AI that should find ways to build in the nuance.

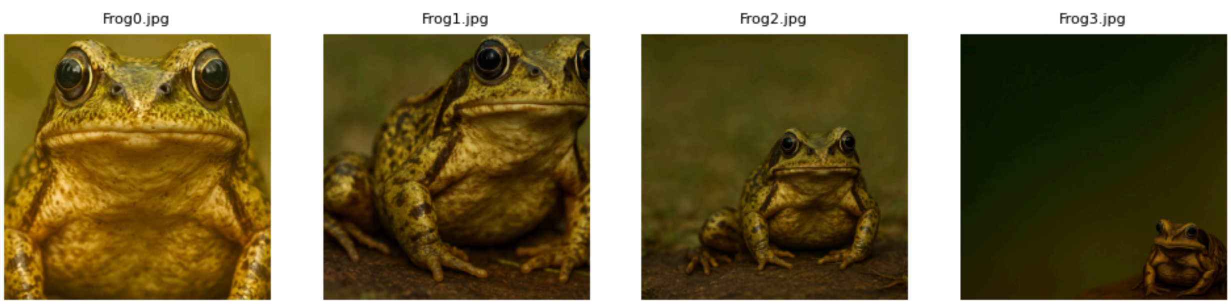
Caveats (expected, not bugs)

- **Minimal-mark abstracts** naturally read **low ρ_r**; that isn’t a failure until it exits the band, but it drags the composite when paired with small Δx.
- **Large clean backgrounds** push **r_v** high; without a counterweight, scores drop even if bands remain OK.
- **Δx in rectilinear work** can stay near-zero if major masses symmetrically straddle the center—even small nudges matter.

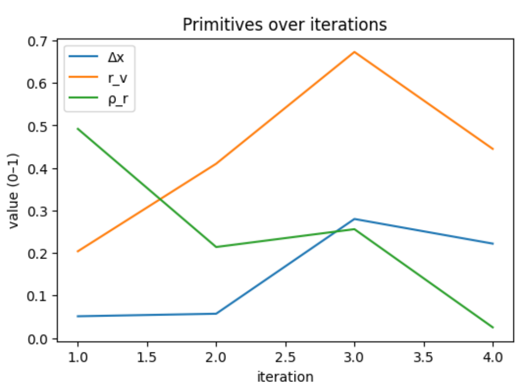
What the Artists Could do Next

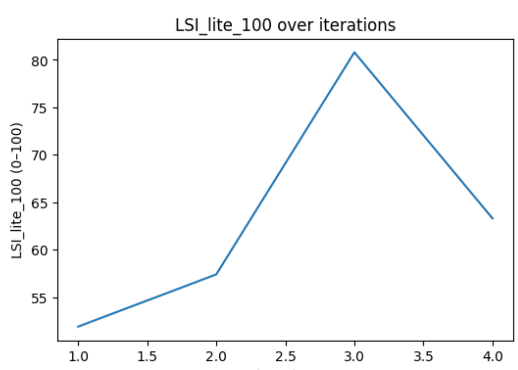
- **Commit to offset:** translate/tilt the stack so **Δx** lands clearly in-band (not near zero).
- **Re-balance the void:** bring **r_v** toward its band center (narrow the blank field or add a counter-shape).
- **Add two decisive edges/overlaps** to lift **ρ_r** slightly without global contrast.
- Re-run a variant of #2 with **Δx↑ + r_v↓ (toward center) + small ρ_r↑**; this combination should clear **55** with all bands OK.

9. Frog



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.051	0.204	0.492	0.519	51.9	OK	OK	OK	FALSE	Frog0.jpg	TRUE	0.296	0.74	increase	TRUE	0.042	0.12	decrease	dx	0.449
1	0.057	0.41	0.214	0.574	57.4	OK	OK	OK	TRUE	Frog1.jpg	TRUE	0.09	0.225	increase	TRUE	0.236	0.674	increase	dx	0.442
2	0.28	0.673	0.256	0.808	80.8	OK	OK	OK	TRUE	Frog2.jpg	TRUE	0.173	0.433	decrease	TRUE	0.194	0.554	increase	dx	0.191
3	0.222	0.445	0.025	0.633	63.3	OK	OK	RED	FALSE	Frog3.jpg	TRUE	0.055	0.137	increase	FALSE	0.425	1	increase	dx	0.256





Results

- **Acceptance:** #0 **False** (51.9) → #1 **True** (57.4) → #2 **True** (80.8) → #3 **False** (63.3).
- **Bands:**
    - **#0:** Δx **OK**, r_v **OK**, ρ_r **OK** → **under gate**.
    - **#1:** all **OK** → **accepted**.
    - **#2:** all **OK** → **accepted** (peak score).
    - **#3:** Δx **OK**, r_v **OK**, $\boldsymbol{\rho_r}$ **RED** → **rejected**.
- **Primitives (0 → 1 → 2 → 3):**
    - $\boldsymbol{\Delta x}$**:** 0.051 → 0.057 → **0.280** → 0.222.
    - $\boldsymbol{r_v}$**:** 0.204 → 0.410 → **0.673** → 0.445.
    - $\boldsymbol{\rho_r}$**:** 0.492 → 0.214 → 0.256 → **0.025**.
- **Priority knob:** $\boldsymbol{\Delta x}$ on all four (scores: 0.449, 0.442, 0.191, 0.256).
- **Directions on card:** increase $\boldsymbol{\Delta x}$ (all), increase $\boldsymbol{r_v}$ early, increase $\boldsymbol{\rho_r}$ throughout.

Researcher's Take

- **Early regime (#0):** all bands OK but the composite (**51.9**) is **below gate**; Δx is small (near-centered), which the Figure-weighting penalizes.
- **Middle regime (#1→#2):** Δx and r_v both rise; with all bands OK the score **climbs sharply**, peaking at **80.8** (#2).
- **Late regime (#3):** $\boldsymbol{\rho_r}$ **collapses to 0.025 (RED)**—likely heavy smoothing or darkness—so the frame **fails** despite decent $\Delta x/r_v$.
- **Limiter history:** knob stays on $\boldsymbol{\Delta x}$ throughout—placement is the persistent constraint; $\boldsymbol{\rho_r}$ becomes the **gate trip** only at #3.

Artist's Take

- #0 reads **center-heavy** and static.
- #1 and especially #2 step back: the frog sits **off-center** with credible breathing room, structure holds and the picture feels more alive.
- #3 loses **edge/texture**; the frog shrinks into a dark, soft corner. Space is fine; **commitment of detail** isn't, hence the reject.

Observations on results (full reality of LSI)

- **Gate precedence:** #0 fails **under gate** (all bands OK); #3 fails because $\boldsymbol{\rho_r}$ **is RED**.
- **Series shape:** score rises as $\boldsymbol{\Delta x}$↑ and $\boldsymbol{r_v}$↑ (#1→#2), then drops when $\boldsymbol{\rho_r}$→**0.025** (#3).
- **Limiter trace: dx → dx → dx → dx** (priority knob) with the **actual gate trip** at #3 driven by ρ_r.
- **Void & placement behaved normally:** r_v stayed in-band and Δx stayed OK after #1; the late failure is **energy-only**, not a measurement artifact.

Plain-English Observations

- The first picture fails because it's **too centered/weak overall**.
- The third picture is the **best**: off-center frog, good use of empty space, enough detail.
- The last picture fails because the **details vanish**—it gets **too smooth/dark**, even though the placement is still fine.

Did the LSI Hold Up?

- **Yes.** It rejected an early **underpowered** frame (#0), accepted when **placement + space** improved (#1–#2), and then correctly rejected when **surface energy collapsed** (#3).

Caveats (expected, not bugs)

- **Low light/denoise** will crater $\boldsymbol{\rho_r}$ (Laplacian energy) even if composition is fine—expected with dark, smoothed backgrounds.
- **Crops shift** $\boldsymbol{\Delta x}$ strongly in close subjects; Figure weighting makes small offsets matter.
- Background uniformity can push $\boldsymbol{r_v}$ high or low without semantics; here it stayed within band and wasn't the blocker.

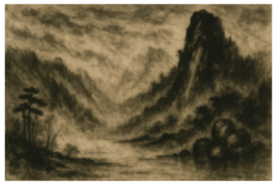
What I'd do next

- **Hold** $\boldsymbol{\Delta x}$ near the #2 placement (clear off-center).
- **Keep** $\boldsymbol{r_v}$ mid-to-high in-band (don't overfill with frog/body; leave a vent).
- **Lift** $\boldsymbol{\rho_r}$ deliberately: two crisp edges (eye ridge + mouth line, or forelimb contour), avoid global blur; add subtle speculars rather than contrast blasts.
- Re-run #2 with **edge accents only**; it should sustain an **accepted** read with margin.

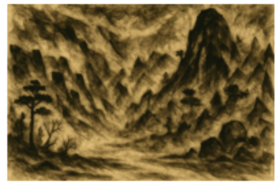
10 Drawing

Drawing0.jpg

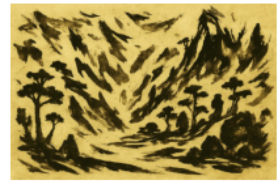
Drawing2.jpg



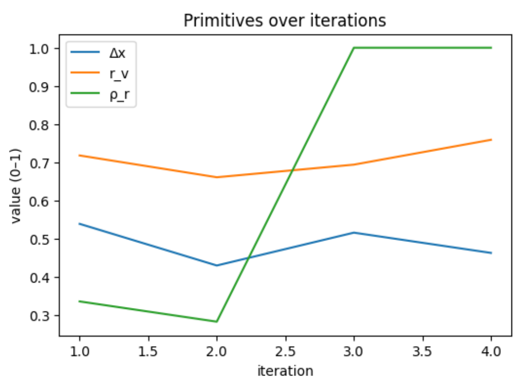
Drawing3.jpg

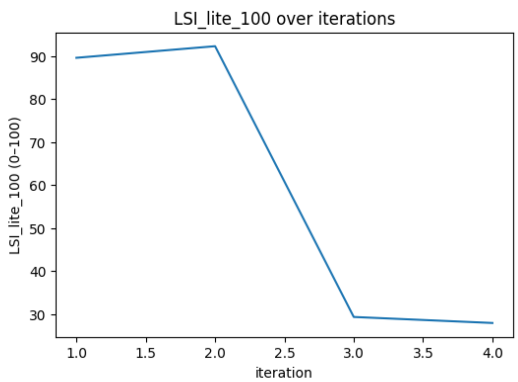


Drawing4.jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score
0	0.539	0.718	0.336	0.896	89.6	OK	OK	OK	TRUE	Drawing0.jpg	TRUE	0.218	0.545	decrease	TRUE	0.044	0.137	increase	rv	0.164
1	0.43	0.661	0.283	0.923	92.3	OK	OK	OK	TRUE	Drawing2.jpg	TRUE	0.161	0.403	decrease	TRUE	0.097	0.303	increase	rv	0.121
2	0.516	0.694	1	0.293	29.3	OK	OK	RED	FALSE	Drawing3.jpg	TRUE	0.194	0.485	decrease	FALSE	0.62	1	decrease	rho	0.3
3	0.463	0.759	1	0.279	27.9	OK	OK	RED	FALSE	Drawing4.jpg	TRUE	0.259	0.648	decrease	FALSE	0.62	1	decrease	rho	0.3





Results

- **Acceptance:** #0 **True** (89.6) → #1 **True** (92.3) → #2 **False** (29.3) → #3 **False** (27.9).
- **Bands:**
  - **#0–#1:** Δx **OK**, r_v **OK**, ρ_r **OK**.
  - **#2–#3:** Δx **OK**, r_v **OK**, ρ_r **RED** ($\rho_r \approx$ **1.00** both frames).
- **Primitives (0 → 1 → 2 → 3):**
  - **Δx:** 0.539 → 0.430 → 0.516 → 0.463 (all healthy, slight drift).
  - **r_v:** 0.718 → 0.661 → 0.684 → **0.759** (high but in-band).
  - **ρ_r:** 0.336 → 0.283 → **1.000** → **1.000** (saturates → out-of-band).
- **Priority knob: r_v** at #0–#1 (decrease), **ρ_r** at #2–#3 (decrease).
- **Nudges on card:** early: **decrease r_v**; late: **decrease ρ_r** (energy); Δx needs only minor attention.

Researcher's Take

- **Early passes** (#0–#1) reflect strong Δx and balanced void/energy.
- **Sharp collapse** occurs when **ρ_r saturates at 1.0** (#2–#3), pushing **band_rho = RED** and driving the composite into the 20s despite Δx/r_v remaining OK.
- **r_v runs high** (0.66–0.76) but never exits its guard; it is **not** the gate tripper.
- **Actionable:** keep Δx where it is, bring **ρ_r down** (less aggressive micro-contrast / mark density in the halo), and nudge **r_v slightly lower** toward its center.

Artist's Take

- The first two drawings breathe and carry a clear off-center mountain mass; paper tone and ink marks feel charged but contained.
- In the last two, the **mark field explodes**, background and subject fuse into a single storm of strokes. Detail turns to **noise**, and the image loses legibility. The mark making reproduction is nice at times, showing observable texture, at times overworked, at times questionable relational placement.

- To regain poise: let the mountain mass carry decision, quiet the background churn, and keep one or two **committed edge seams** instead of universal texture.

Observations on results (full reality of LSI)

- **Gate precedence:** #2–#3 fail because **ρ_r is RED** (≈1.00), regardless of the otherwise healthy $\Delta x/r_v$.
- **Series shape:** Δx stays serviceable; **r_v climbs but stays in-band**; the **ρ_r spike** alone explains the cliff-drop from ~90 to ~28–29.
- **Limiter history: rv → rv → rho → rho**—void was the gentle limiter early; energy became the hard stop late.
- **No measurement artifact indicated:** ρ_r's rise matches the visible surge in small-scale contrast/texture; the failure is structural (energy overload), not a meter glitch.

Plain-English Observations

- The first two pictures pass: clear placement, enough space, texture under control.
- The last two fail because the **texture gets too wild**—there's **too much small, sharp detail everywhere**.
- Placement and space stay fine; it's the **over-energized surface** that breaks the test.

Did the LSI Hold Up?

- **Yes.** It accepted the balanced drawings, then rejected when **surface energy blew past the allowed range**, even though placement and space remained okay—exactly how the band + gate rule is supposed to behave.

Caveats (expected, not bugs)

- **Scan sharpening / denoise + sharpen** can spike ρ_r to near-1.0, especially on textured paper.
- **Charcoal/ink crosshatching** that fills the halo region will read as **excess energy** by the Laplacian measure.
- **High r_v** in paper-heavy compositions is normal; unless it crosses the guard, it won't trigger the gate.

What I'd do next

- **Reduce ρ_r** without dulling the focal mass: ease micro-contrast in the **halo** (background field), keep **two decisive edges** on the mountain ridge/foreground tree, and let the rest soften.
- **Trim r_v** slightly toward center (tighten the paper field or darken a vent) while preserving the off-center mass (**Δx**).
- If the intent is truly **expressionist mark density**, consider evaluating with the **MarkMaking_Expressive** profile (more permissive ρ_r band) for a second read—then compare acceptance across profiles.

Interim Summary: what the first ten sets show

Big picture

- **LSI held up.** We saw both expected regimes: strict **band trips** (Δx RED, r_v RED, ρ_r RED) and **under-gate** rejections when structure was in-band but weak overall.
- **Acceptance so far: 14 / 37 frames (≈38%)** accepted across all sets—healthy for a structural gate meant to surface limits, not crown winners.

Recurrent limiters (by axis)

- **Δx (placement):** Most common early limiter. It tripped or suppressed scores in **Portrait**, **Walking**, **Abstract**, and **Collapse**. When Δx improved, other axes took over as the bottleneck.
- **r_v (void):** Two distinct behaviors:
  - **True void collapse** (scene packed/filled) → **Still life #3**, **Abstract #2**.
  - **Mask-as-subject** in interiors → **Room** (walls/floor labeled foreground → r_v RED all runs).
- **ρ_r (energy):** Both directions showed up:
  - **Energy collapse** (too smooth/dark) → **Frog #3**, **Walking #2–3** (dragging the composite).
  - **Energy overload** (texture everywhere) → **Drawing #2–3** ($\rho_r \approx 1.0$, RED).

Profile effects (what the meter weights)

- **Figure_Default: Δx** carries the most weight; modest placement changes move the needle.
- **Landscape: r_v** weight dominates; scores fell when openness collapsed even as Δx improved (Set 2).
- **MarkMaking-style work:** We didn't run the permissive profile, but the **Drawing** overload suggests when to consider it for comparison.

Per-set one-liners

- **Collapse (tower):** Pass → peak → drift → **fail on Δx RED** (center creep).
- **Landscape:** All **pass**; score dip maps to **r_v shrink** despite Δx gains.
- **Portrait:** ρ_r RED → under-gate → **pass** once **Δx** commits.
- **Room: r_v RED** throughout (mask makes room the subject).
- **Still life:** Mid-series looks good but **ρ_r RED** (too soft), final also **r_v RED** (space closed).
- **Figure (charcoal):** Pass → pass → **under-gate** with over-open field → marginal **pass**; Δx is the persistent knob.
- **Walking:** Δx RED early; later **under-gate** with low ρ_r—none pass.
- **Abstract:** Δx RED → Δx OK but **over-open + low ρ_r** → under-gate—none pass.
- **Frog:** Under-gate → **pass** → **strong pass** → **ρ_r RED** (too smooth/dark).

- **Drawing (landscape ink):** Two clean **passes**, then **ρ_r overload** (mark storm) → two fails.

What consistently worked

- **Fix placement first** (Δx) for figures and close subjects.
- **Keep r_v near band center**—either open a vent or prevent seal-up, but avoid extremes.
- **Raise or trim ρ_r with **two committed edges** or by quieting noisy halos; avoid global sharpen/blur.

Expected caveats (not bugs)

- **Interiors:** non-semantic masks can label the room as subject → low r_v. (needs to be further tested, this image set arguably, the room was the subject)
- **Painterly/soft light:** lowers ρ_r; fine until it exits the band, but it drags scores near the gate.
- **Scan/processing:** denoise/sharpen can artificially crater or spike ρ_r.
- **Small crops:** swing Δx more than intuition suggests (weighting makes this visible).

Next additions (extend the study)

- Run **paired variants** that isolate a single knob (Δx-only, r_v-only, ρ_r-only) to produce clean cause→effect plots.
- For dense mark fields, **compare profiles** (Figure_Default vs MarkMaking_Expressive) to separate "overload" from "intentional texture."
- Add a brief **subject-mask note** per interior/interior-like set to document the chosen stance (room-as-figure vs assemblage).

TL;DR

LSI-lite is doing its job: it reliably flags **centering drift**, **void collapse/over-openness**, and **energy collapse/overload** across very different images. The main friction isn't math; it's **subject definition** (r_v) and **profile fit**. Treat it as **telemetry you can steer by**, not a judge.

Concerns (watch-outs)

1. **Non-semantic r_v.** Interiors and tone-flat scenes can be read as "room-as-subject," making r_v look bad even when the photo feels spacious. You'll need a stance (default vs. override mask).
2. **Δx brittleness.** Small crops, micro shifts, or pose tweaks swing Δx a lot (especially in Figure); easy to misinterpret as "composition change."
3. **ρ_r sensitivity to processing.** Denoise/blur tanks it; scan sharpen spikes it. Logging capture/processing is mandatory to compare runs.
4. **Profile mismatch.** Dense mark fields fail under Figure but might be fine under MarkMaking_Expressive. Without an explicit profile choice, you'll mislabel intent.
5. **Gate strictness.** Teams will fixate on "≥55" and ignore band violations; the gate is two-part by design, expect arguments, but gotta start somewhere.
6. **Gaming risk.** It can "pass" by nudging gauges toward centers without making the picture better. Keep the **Artist's (human) Take** in the loop.
7. **Environment drift.** Different OpenCV/Laplacian/Canny defaults shift numbers slightly; pin or record versions when publishing comparisons.
8. **Scope limits.** No sense of semantics, hierarchy, or story. It measures structural behavior only.

Opportunities (use it to win)

1. **Prompt A/B at scale.** Run Baseline→Pressure→Contradiction sets; log which knob fails first—great for model comparisons.
2. **Guardrail for generators.** Reject near-centered portraits (Δx RED) or over-smooth outputs (ρ_r RED) **before** shipping to users.
3. **Failure atlas.** Cluster images by first failing knob (dx/rv/rho) to discover model-specific weaknesses; feed back into training or prompting.
4. **Profile dual-read.** Score tricky sets under **Figure_Default** and **MarkMaking**; disagreements are real signal, not noise.
5. **Mask override option.** For interiors/products, allow a subject mask input; r_v becomes aligned with intent without changing the metric.
6. **Series telemetry.** Treat trajectories (e.g., $\Delta x\uparrow$, $r_v\downarrow$, $\rho_r \rightarrow$low) as fingerprints of collapse or re-cohere behavior, useful for engine regression tests.
7. **Human-in-the-loop critique.** Pair the read card with a two-edge prescription ("commit edges X and Y"), fast, teachable fixes for artists.
8. **Dataset curation.** Filter training/eval sets by structural variety (Δx and r_v coverage) to avoid over-centered, sealed datasets.
9. **Reward shaping.** Use in RL/fine-tune loops as a soft reward (stay within bands while meeting task constraints).

10. **Dashboards.** Ship a simple UI: band gauges, acceptance stamp, and a “priority knob” nudge; it’s enough for daily decisions.

Real-world playbook (do this next)

- **Lock the stance on r_v.** Default = Otsu/largest-component; allow an **optional mask override** for interiors/product shots. Document it.
- **Two-profile policy.** If mark density is the point, always run **Figure + MarkMaking** and show both stamps.
- **Pin versions + capture metadata.** Save OpenCV/Numpy versions and processing notes per run.
- **Write two prescriptions per fail.** For each RED or under-gate: (a) one **placement/void** move, (b) one **edge/energy** move.
- **Baseline suite.** Keep 8–10 canonical prompts/images and re-run after any model or workflow change to catch drift.

Where it already pays off (from the sets)

- **Δx drift caught** (Collapse, Portrait, Walking, Abstract).
- **Void stance exposed** (Room)—not a bug; a choice.
- **Energy collapse/overload named** (Frog: collapse; Drawing: overload).
- **Landscape weighting validated** (r_v drives score more than Δx).

Bottom line: the metric is sturdy enough for **production telemetry** and **research baselining**. The biggest unlock is a **clear r_v policy** and consistent **profile selection**; after that, it becomes a sharp tool for both engine teams and artists.